```
!pip install plotly

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

sns.set(style="darkgrid")
```

```
Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packages (5.24.1)
Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from
```

```
from google.colab import files
uploaded = files.upload()
```

Choose Files survey.csv

**survey.csv**(text/csv) - 303684 bytes, last modified: 3/25/2026 - 100% done

```
Saving survey.csv to survey.csv
```

```
df = pd.read_csv("survey.csv")
df.head()
```

| | Timestamp | Age | Gender | Country | state | self_employed | family_history | treatment | work_ |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 2014-08-27 11:29:31 | 37 | Female | United States | IL | NaN | No | Yes | |
| **1** | 2014-08-27 11:29:37 | 44 | M | United States | IN | NaN | No | No | |
| **2** | 2014-08-27 11:29:44 | 32 | Male | Canada | NaN | NaN | No | No | |
| **3** | 2014-08-27 11:29:46 | 31 | Male | United Kingdom | NaN | NaN | Yes | Yes | |
| **4** | 2014-08-27 11:30:22 | 31 | Male | United States | TX | NaN | No | No | |

5 rows × 27 columns

```
# Drop irrelevant columns
df = df.drop(['comments', 'state'], axis=1, errors='ignore')

# Rename columns for simplicity
```

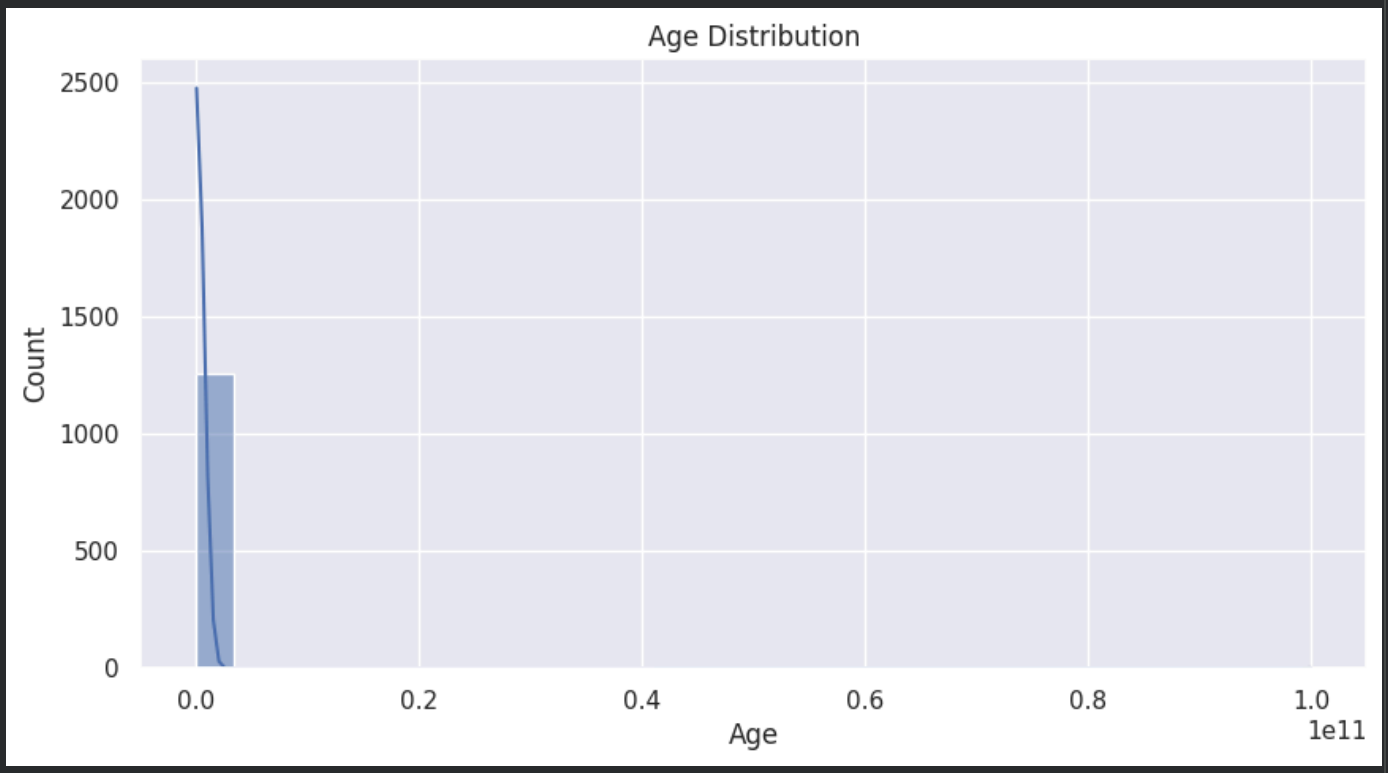
```
df.columns = df.columns.str.strip().str.replace(' ', '_')

# Handle missing values
df['self_employed'].fillna('No', inplace=True)
df['work_interfere'].fillna('Dont know', inplace=True)

# Clean Gender column (important step)
def clean_gender(g):
    g = str(g).lower()
    if 'male' in g:
        return 'Male'
    elif 'female' in g:
        return 'Female'
    else:
        return 'Other'

df['Gender'] = df['Gender'].apply(clean_gender)

df.head()
```

```
/tmp/ipykernel_5186/341665074.py:8: FutureWarning: A value is trying to be set on a copy o
The behavior will change in pandas 3.0. This inplace method will never work because the in

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col:


  df['self_employed'].fillna('No', inplace=True)
/tmp/ipykernel_5186/341665074.py:9: FutureWarning: A value is trying to be set on a copy o
The behavior will change in pandas 3.0. This inplace method will never work because the in

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col:


  df['work_interfere'].fillna('Dont know', inplace=True)
```

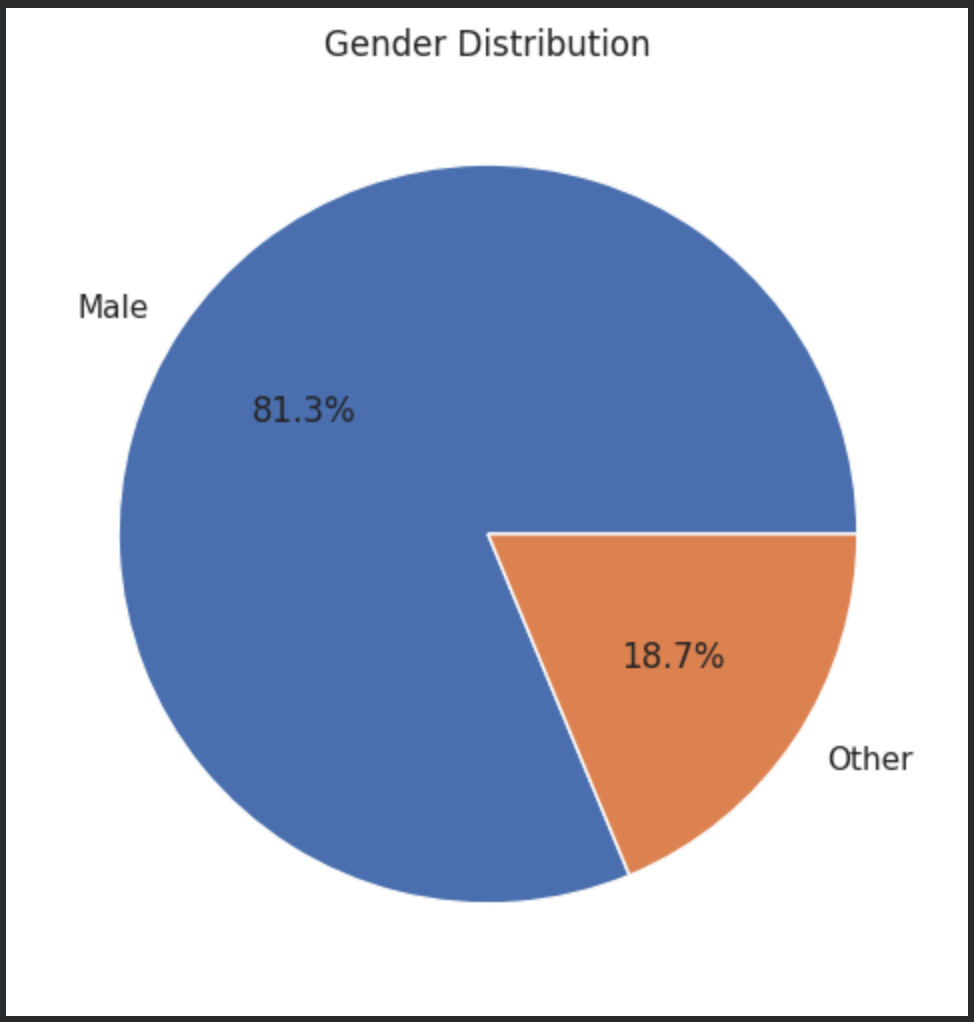
| | Timestamp | Age | Gender | Country | self_employed | family_history | treatment | work_interfe |
|---|---|---|---|---|---|---|---|---|
| **0** | 2014-08-27 11:29:31 | 37 | Male | United States | No | No | Yes | Off |
| **1** | 2014-08-27 11:29:37 | 44 | Other | United States | No | No | No | Rar |
| **2** | 2014-08-27 11:29:44 | 32 | Male | Canada | No | No | No | Rar |
| **3** | 2014-08-27 11:29:46 | 31 | Male | United Kingdom | No | Yes | Yes | Off |
| **4** | 2014-08-27 11:30:22 | 31 | Male | United States | No | No | No | Ne |

5 rows × 25 columns

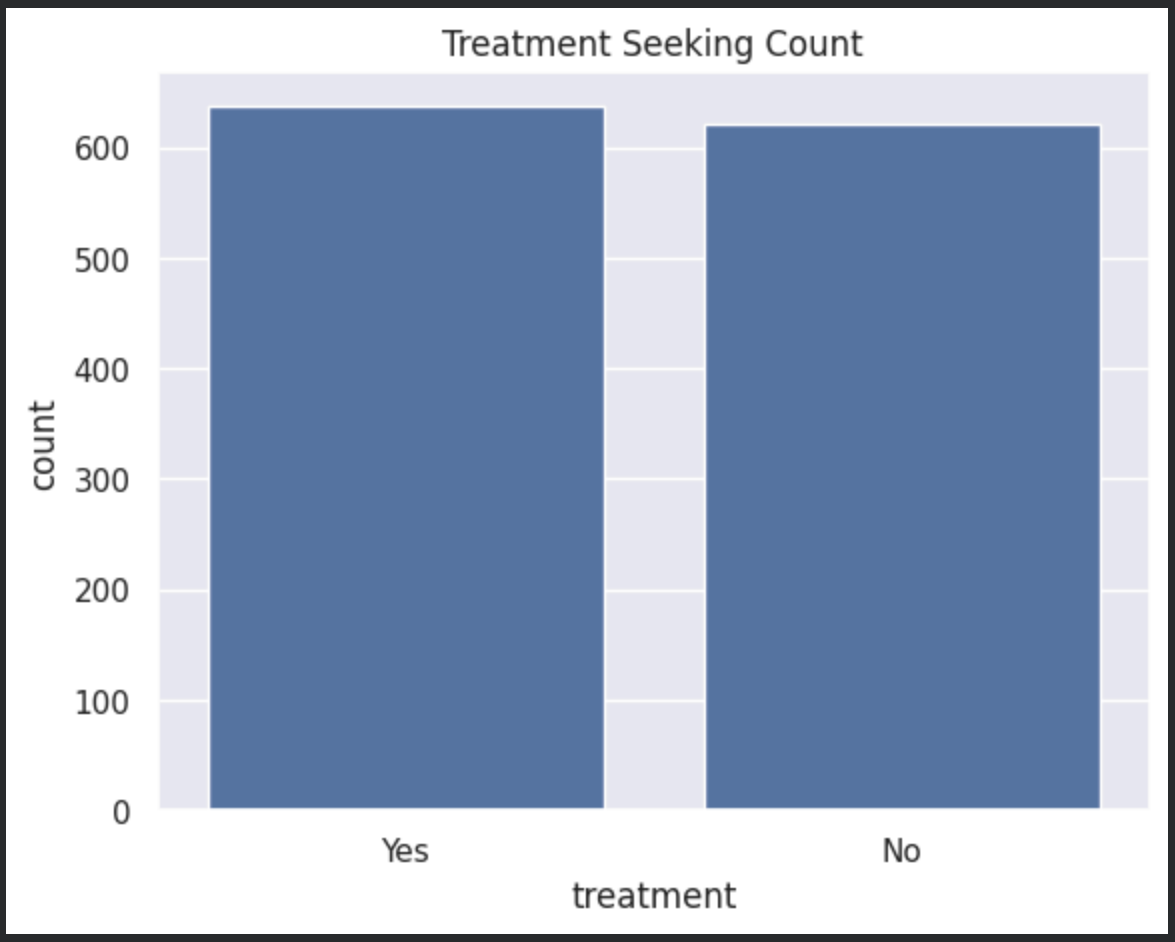
```
plt.figure(figsize=(10,5))
sns.histplot(df['Age'], bins=30, kde=True)
plt.title("Age Distribution")
plt.show()
```



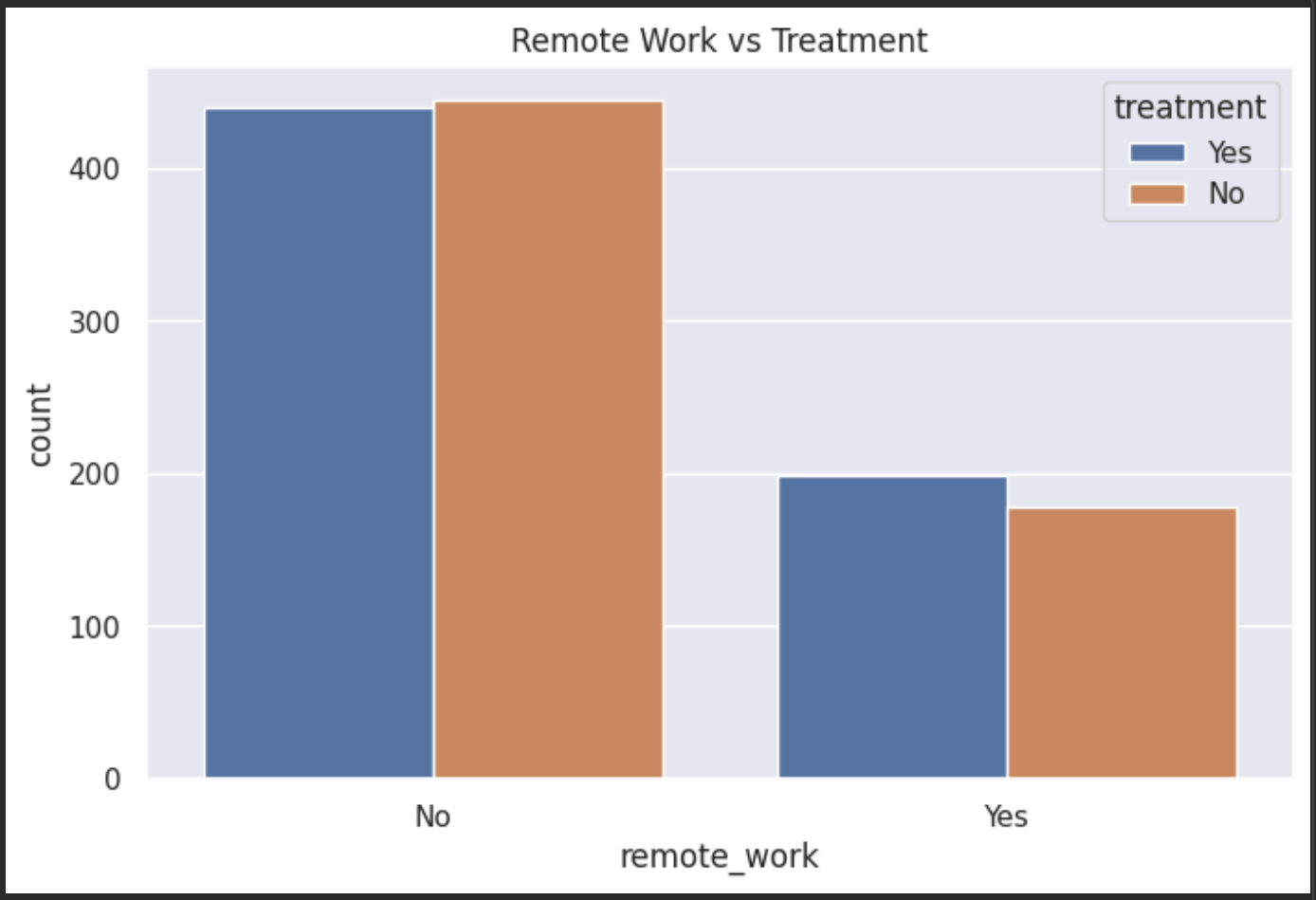

```
df['Gender'].value_counts().plot.pie(autopct='%1.1f%%', figsize=(6,6))
plt.title("Gender Distribution")
plt.ylabel("")
plt.show()
```



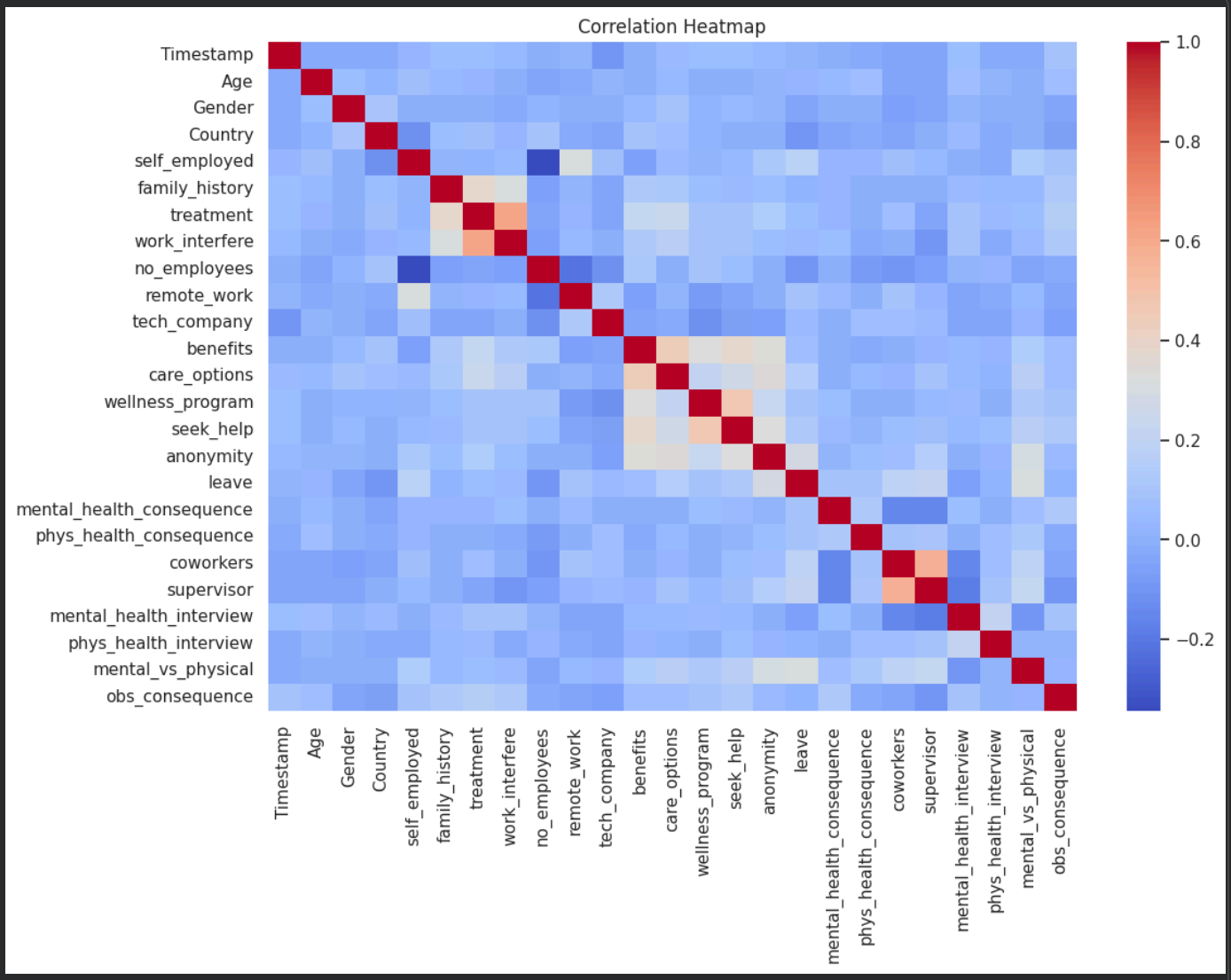

```
sns.countplot(x='treatment', data=df)
plt.title("Treatment Seeking Count")
plt.show()
```



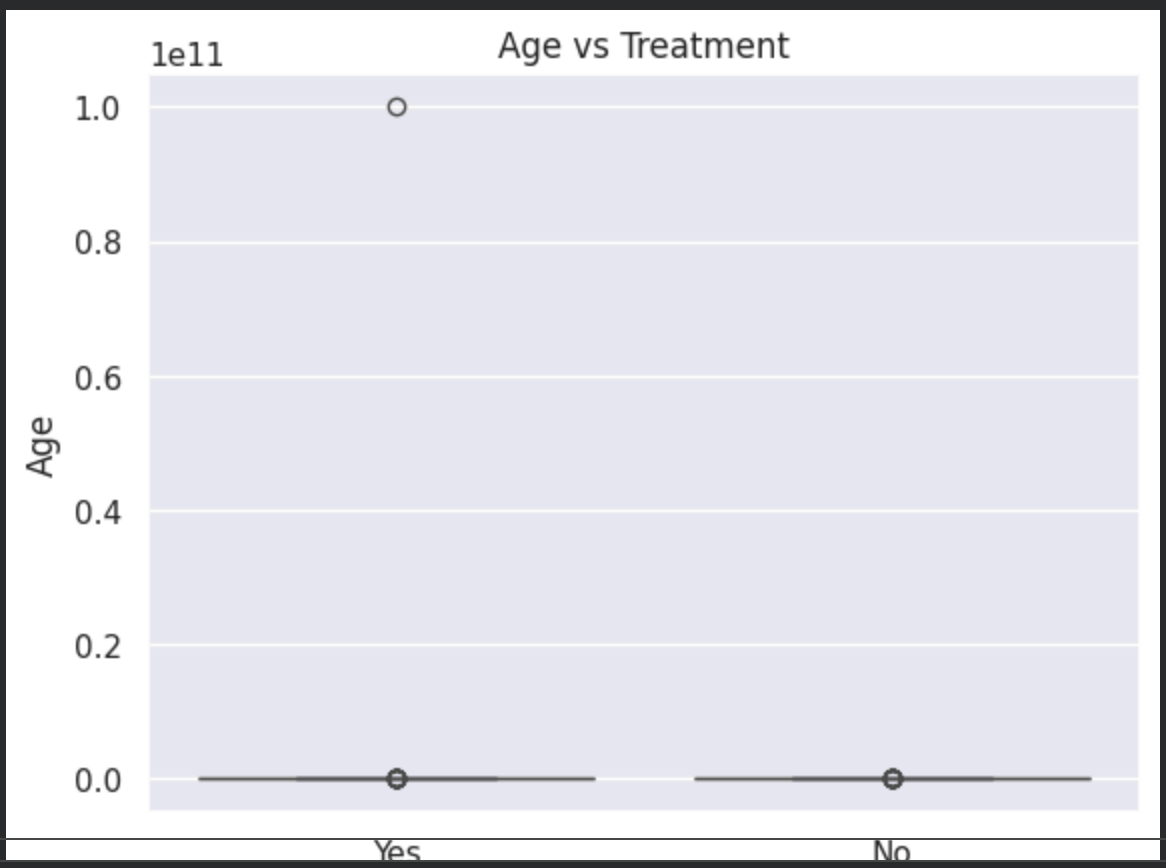

```
plt.figure(figsize=(8,5))
sns.countplot(x='remote_work', hue='treatment', data=df)
plt.title("Remote Work vs Treatment")
plt.show()
```

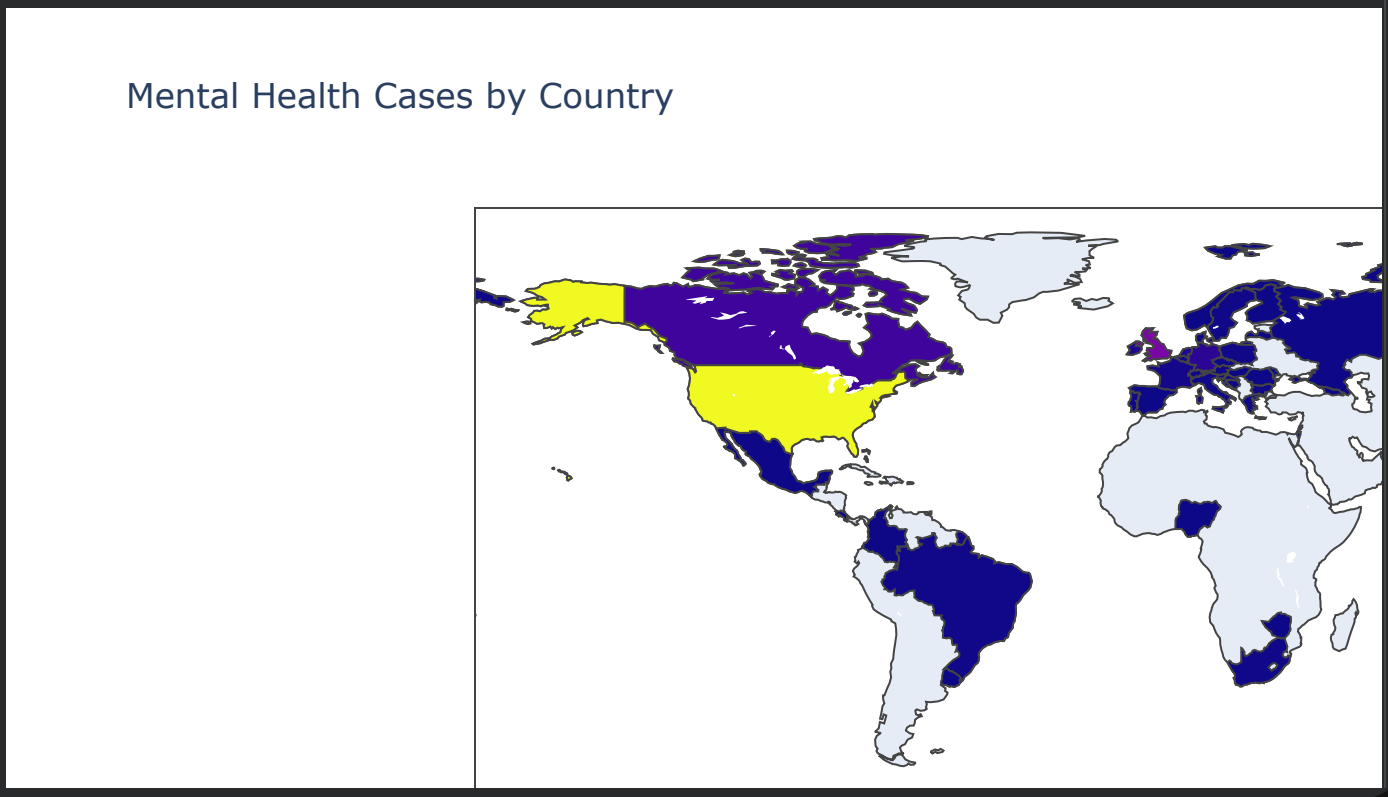



```
# Convert categorical to numeric
df_encoded = df.copy()
for col in df_encoded.select_dtypes(include='object').columns:
    df_encoded[col] = df_encoded[col].astype('category').cat.codes

plt.figure(figsize=(12,8))
sns.heatmap(df_encoded.corr(), cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```




```
sns.boxplot(x='treatment', y='Age', data=df)
plt.title("Age vs Treatment")
plt.show()
```




```
country_data = df.groupby('Country')['treatment'].count().reset_index()

fig = px.choropleth(country_data,
                    locations='Country',
                    locationmode='country names',
                    color='treatment',
                    title="Mental Health Cases by Country")

fig.show()
```